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Machine Learning in PV Fault Detection, Diagnostics and Prognostics: A Review

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***Abstract* — Photovoltaic (PV) system malfunctions cause output efficiency to lower which consequently lowers the return of the investment (ROI) and delays investment payback times. These malfunctions can be limited by implementing Photovoltaic System Monitoring (PVSM) solutions. Recently, Machine Learning Techniques (MLT) have been implemented to improve PVSM results and aid in PV performance and PV fault detection, identification, diagnostics and prognostics. This paper provides a review of the work done in the MLT PVSM research field, provides an organized list of MLT solutions used in PVSM, and provides a list of opportunities and challenges to further research in the PVSM field.**

***Index Terms* — Diagnostics, Machine Learning, Prognostics, PV Fault, PV System Monitoring, Return of the Investment.**

I. Introduction

Over the past few years, photovoltaic (PV) systems have become very popular amongst homeowners and solar plant investors since they are viewed as low risk investments that can achieve high return rates and attractive investment payback times [1].

The PV systems are composed by photovoltaic modules and the balance of system (BOS) that includes wiring, switches, a mounting system and solar inverters. In order to avoid PV system downtime and maintain the output efficiency at high levels to shorten investment payback periods, it is essential for the investors, owners and maintenance operators to have access to the input and output data of the PV system. This data can only be acquired through monitoring systems and should be accessible in real-time for the PV system faults to be addressed accordingly to avoid production loss. The PV system real-time information is useful to continuously keep investors and owners informed about the Return on the Investment (ROI), and keep the maintenance operators aware of the PV system status to easily schedule maintenance trips.

Research on PV system Fault Detection and Diagnostics (FDD) has made it possible to detect PV system faults and correctly diagnose them when using Machine Learning Techniques (MLT). Research on MLT is popular and has become very accurate due to available software and computing capacity of the new computers. The two main model categories used in MLT are classification and regression. Classification is used to detect and identify PV faults while regression is used in PV system diagnostics which allows for PV system performance analysis or solar output estimations.

Alongside research on PV system FDD which predicts current faults, there is also ongoing research on Fault Prognostics that predict the performance over the remaining useful life (RUL) of the PV system and consequently predicting possible future faults by using regression MLT. The RUL of a PV system is determined by estimating the degradation rate of the PV modules through degradation models that consider UV radiation, temperature, humidity, state-of-health models and end of life models [2], [3], [4].

By combining PV system FDD and Fault Prognostics research areas it is possible to develop PV system Prognostics and Health Management (PHM) [4] approaches which helps to plan preventive, corrective and/or condition-based maintenance trips contributing to lower the O&M costs by an estimated 20% and consequently shorten investment payback times on the ROI [5]. Condition-based maintenance is the new term associated to the prognostics approach, in which the faults are predicted and the condition-based maintenance trips are made when the faults occur and therefore lowers the frequency of preventative measures and reduces the impacts and costs of the corrective measures by anticipating failures or catching them early [6]. Fig. 1 illustrates how the Machine Learning Techniques are used in the PVSM field of research.

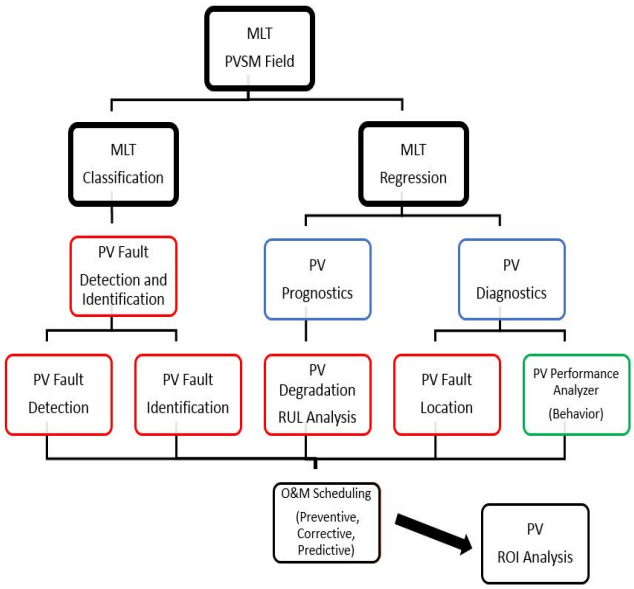


Fig. 1. MLT in the PVSM field of research

According to the thesis written by Zhao, PV system faults may remain hidden due to the connection infrastructure of the PV system and MLT can successfully detect and identify these types of faults, also known as "blind-spots" [7]. Thus, demonstrating the importance of the use of data-driven models or MLT in PV fault detection, diagnostics and prognostics over model-based techniques. Data-driven approaches are also recommended to deal with non-linear, unpredictable and complex settings such as the PV system solar production that deal with incoherent weather conditions that impact the solar production of the PV system.

The Machine Learning research area is vast, but recently Pedro Domingos shared in his book [8] (recommended by Bill Gates) that MLTs can be organized into five "tribes" and that each one is related to a specific master algorithm as shown in Table I. This concept helps to organize and narrow down the MLTs that are used in PV System Monitoring (PVSM) and indicate which MLTs that are not being used and may have the potential to present good results in the PVSM research field. Pedro Domingos defends the idea of combining various master algorithms presented in Table I to make one master algorithm. This idea of combining different master algorithms is already being adopted in the PVSM research field but in a smaller scale by combining two or three Master Algorithms and are referred to as hybrid MLTs. Two examples of hybrid MLT can be found in [9] and [10] where the first analysis a hybrid MLT that uses the Genetic Algorithm and an Artificial Neural Network which combines the master algorithms of tribes 3 and 2. The latter work tests a hybrid MLT that uses a sparse Bayesian learning theory and Support Vector Machines which combines the master algorithms of tribes 4 and 5.

TABLE I
MLT TRIBES AND RESPECTIVE MASTER ALGORITHMS [8]

	Tribe	Master Algorithm
Tribe 1	Symbolists	Inverse deduction / Induction
Tribe 2	Connectionists	Backpropagation
Tribe 3	Evolutionaries	Genetic Programming
Tribe 4	Bayesians	Probabilistic Inference
Tribe 5	Analogizers	Kernel Machines

Based on this information, the motivation was to understand how the different MLT have been used in the different research areas of the PVSM field such as PV Fault Detection, Identification, Diagnostics and Prognostics. The main goal of this work is to analyze 90 research papers and make a list of the MLT and organize them into their associated MLT tribe. The other goal of this work is to present the number of experiments associated to a parameter (PV system faults, PV system scale, Hybrid MLTs, etc.) related to the MLT PVSM field. This experiment count provides information about the areas of the PVSM research field that are more popular as well as those that show good results but need more research. The challenges and opportunities that are available in the PVSM field are also provided in this work.

After this introduction section, a methodology description is provided in section II. The results are presented and discussed in section III, which is then followed by a conclusion of the work in section IV.

II. METHODOLOGY

This section describes the methodology that was applied, in order to understand which MLTs are being studied in the PV system monitoring (PVSM) research field and which should be studied further. First, a total of 90 scientific papers were identified as being related to the MLT PVSM research field. Secondly, an experiment count was made according to specific parameters. These parameters are namely the type of:

- MLT tribe (Tribe 1, Tribe 2, Tribe 3, Tribe 4, Tribe 5);
- PVSM technique (Model-based, Data-Driven, Hybrid MLT);
- MLT models (Classification, Regression);
- PVSM method (Detection, Identification, Diagnostics, Prognostics);
- PVSM papers (PV performance, PV fault);
- PV system testbed (Grid PV system, lab PV system, simulation); In this paper, a grid PV system is assumed to have more than 3kWp of installed power which represents a typical roof-top household PV system.

In addition, the experiment count is also applied to the parameters that do not involve MLT and that influence the PVSM research analysis such as the type of

- PV system faults (arc, short-circuit, faulty condition, degradation, shading, Line to Line, soiling, open-circuit);
- PV system scale (roof, commercial, plant);
- PVSM level (array, string, module);
- PV system measured inputs (irradiance, module temperature, ambient temperature, current, voltage, etc.) and outputs (power, energy, I-V curve, decision).

The following four subsections present a brief description of all the PVSM techniques that were studied in this work which include the model-based approach, data-driven regressive approach, machine learning techniques and the hybrid MLTs.

A. Brief Description of the Model-Based Approach

The Model-Based (MB) approach in PVSM is achieved by using sensor data associated to mathematical and visual methods that have indicators and thresholds to aid in decision making [11]. Some examples of this type of approach identified during this work include the **Persistence Model**, and the **Physical Model**.

The Persistence model may be used as a forecasting technique which assumes that the weather conditions remain the same between one sample of the time-series and the next one. The Physical model consists of a set of equations that describe the behavior based on the physical principles. This model may consist of a sensor and stored data which is then analyzed by indicators and thresholds that were determined through trial and error experiments [11].

B. Brief Description of the Data-Driven Regressive Approach

The Data-Driven Regressive (DDR) approach in PVSM includes all the data-driven algorithms that are not learning algorithms but are regressive algorithms and depend on historical data for forecasting [11]. The most popular DDR algorithms that were identified in this work are the **ARIMA** (Auto-Regressive Integrated Moving Average), the **linear regression models** and the **Principal component analysis** (PCA).

The forecasting values of ARIMA are based on its inertia, while the regression models analyses the relationship between the dependent and independent variables. The PCA is used to select the most relevant variables [11].

C. Brief Description of the Machine Learning Techniques

This section briefly describes the MLT Tribes and indicates which of the MLT are most popular in each tribe. Table I presents a description of the five tribes that follow.

Tribe 1: Symbolists (Inverse deduction / Induction)

The MLTs that belong to this tribe attempt to find the missing knowledge by applying inverse deduction which starts with some properties and conclusions and works its way backward to find the gaps in knowledge by using analysis and data sets [8]. The most popular MLT of this tribe are Decision Trees, Random Forests and the Fuzzy Logic [11].

Tribe 2: Connectionists (Backpropagation)

The MLT of this tribe are composed by algorithms that claim to emulate the functions of the brain by creating artificial neurons and connecting them in a neural network by using an input layer, one or various hidden layers and an output layer. The inputs are taken by the hidden neurons which then generate an output (new knowledge) that can be read by other neurons that perform the same function [8]. The Artificial Neural Networks and Extreme Learning Machine are some the most popular tribe 2 MLTs [11].

Tribe 3: Evolutionaries (Genetic Programming)

Here the algorithms of the MLTs try to mimic the evolutionary process of genomes and DNA where the performance is measured by the fitness of the offspring. The algorithms are divided into a set of sub programs that mutate and combine with other sub programs to determine if the prediction results improve with multiple iterations of the program. New knowledge is determined by simulating evolution [8]. The popular evolutionary MLTs include the Genetic Algorithm and the Genetic Programming [11].

Tribe 4: Bayesians (Probabilistic Inference)

These MLTs systematically attempt to reduce the uncertainty of the new knowledge by applying the probabilistic inference algorithm to events. The events that are known to occur are assigned prior probabilities, and as more evidence is observed, the priors either lose their importance or become more important [8]. The Monte Carlo Method is a very popular Bayesian MLT [11].

Tribe 5: Analogizers (Kernel Machines)

The Analogizers MLTs recognize the similarities between the old and the new data/knowledge by using the nearest neighbour kernel machine algorithms which analyse its surroundings and try to generalize by similarity. These results are similar to those of the neural network models [8]. The most popular analogizer MLTs are the Support Vector Machines and the K-Nearest Neighbour algorithms [11].

D. Brief Description of the Hybrid MLTs

Hybrid MLTs is the name given to the combination between two or more PVSM techniques (MB, DDR, and MLTs). This combination contributes to strengthening the accuracy of the forecasting results [11]. The most popular hybrid MLTs that were identified in this work are namely the **Radial Basis Function Neural Network (RBFNN),** the **ANN + Physical Model,** the **Genetic Algorithm-Artificial Neural Network** (**GA-ANN),** the **ANFIS (adaptive neuro-fuzzy inference system),** and the **Relevance Vector Machine (RVM).**

The RBFNN is a MLT combination between tribe 5 (RBF) and tribe 2 (NN). The ANN+Physical model is a PVSM technique combination between a tribe 2 (ANN) MLT and a model-based technique. The GA-ANN is a MLT combination between tribe 3 (GA) and tribe 2 (ANN). The ANFIS is another MLT that combines tribe 2 (neuro) and tribe 1 (fuzzy).

III. Results and Discussion

Over 150 PV systems monitoring (PVSM) research papers were identified and analysed for this work. Only 90 of them were considered since they include the topics that are discussed in this work such as MLTs, Hybrid MLTs, power and energy estimation, as well as PV performance and PV fault analysis. The papers that only present the framework and do not include an experimental research were not considered. The graphs that follow present the number of times a given parameter is mentioned in the 90 papers that were studied (blue). The graphs also present the number of times that a given parameter is mentioned in the 32 papers that study the PV fault analysis (orange). Some of these papers focus only on the PV performance analysis, others only on the PV fault analysis and others on both the PV performance and PV fault analysis.

Fig. 2 presents the various MLT tribes and how many times they were used in all the papers and in the PV fault papers. This graph also presents the experiment count of the Model-Based, Data-Driven Regressive and hybrid MLT approaches. The MLTs of Tribe 2 and 3 are used the most in the PVSM research field which include the ANN and SVR respectively. The Regression Trees and the Bayesian probability from the Tribes 1 and 4 respectively are used similarly. The MLTs from Tribe 3 (Genetic Programming and Genetic Algorithm) are the least used and not used at all in the PV fault papers. Fig. 2 also presents the experiment count that use the three

approaches namely the MB approach, the DDR approach and the Hybrid MLT approach. The hybrid MLT approach is used more in the PV performance analysis compared to the PV fault analysis.

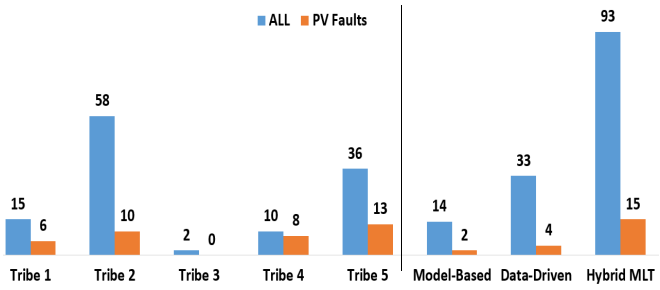


Fig. 2. MLT Tribes and PVSM Technique approaches

Fig. 3 presents the number count of the papers for the classification and regression MLT models as well as the four PV system monitoring methods such as the Detection, Identification, Diagnosis and Prognosis. Fig. 3 also presents the number count of the papers that analyse the PV performance as well as those that analyse the PV faults. There are double the papers that consider regression as opposed to classification MLT models in the PV system monitoring field of research. Classification is mostly used in the PV fault analysis. PV fault Detection is the most popular PVSM method and prognosis the least popular, while PV fault identification and diagnosis are similar in use. We can see that PV performance analysis was used in 87 out of the 90 papers studied in this work while 29 of the 32 PV fault papers studied in this work performed PV performance analysis. PV performance research is more popular than PV fault analysis.

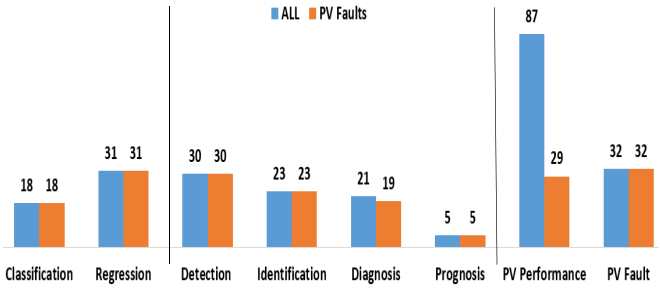


Fig. 3. MLT Models, Methods and PV System Analysis

Fig. 4 presents the parameters of the testbed used in the research papers. The end results of the experiments are always more reliable when using a grid connected PV system. From the graph below, it is possible to see that there are many experiments carried out on PV grid connected systems. However, there are still a fair amount of experiments that are carried out in lab testbeds (a few PV modules) and by simulating a PV system using software (Matlab). These two latter testbeds are not as reliable because the nonlinear nature of the weather conditions may not be considered in the analysis. Models always have limitations compared to the true PV systems. The PV fault analysis research is mainly done on lab and simulation testbeds which might not result in reliable findings. The PV installed power is also reflected in the graph that follows by differentiating the PV scale into plant, commercial and roof-top sizes. There is an even amount of work that is carried out in all three scales. The monitoring level of the PV systems is normally made on the array level due to cost control, however it is interesting to see that the PV fault analysis research tends to use the module-level monitoring which in turn provides very accurate results.

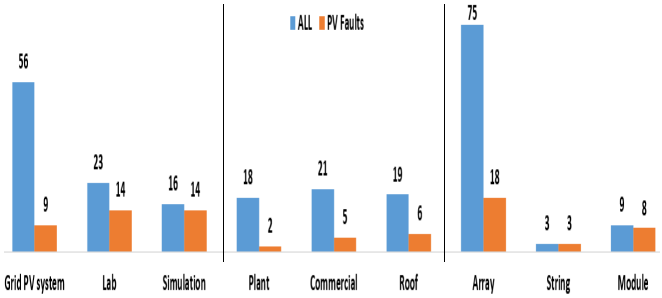


Fig. 4. PV System Testbeds, Scale and Monitoring Level

Fig. 5 presents the input features that are most often used in the research papers studied in this work. Irradiance, ambient temperature and power are the input features that are used the most followed by the wind speed, module temperature, current and voltage values. Not many papers consider the energy values.

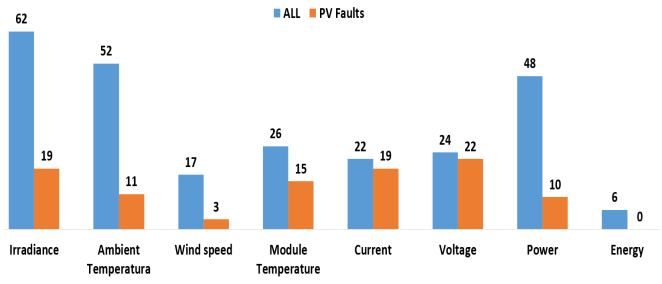


Fig. 5. PVSM Input Features

Fig. 6 presents the output features that are mostly used in the research papers studied in this work. Power prediction is the most popular while energy and I-V curve prediction is also used. Classification decisions are widely used in the PV fault analysis as well as the power and I-V curve predictions. Not many papers consider the standalone voltage and current predictions.

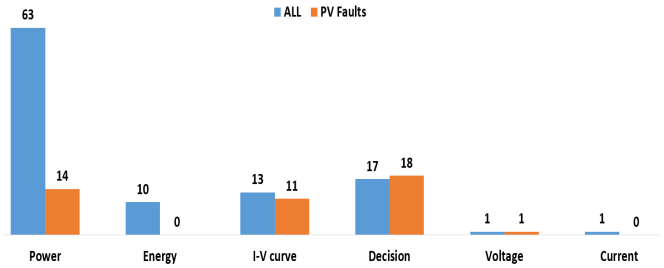


Fig. 6. PVSM Output Features

Fig. 7 presents the experiment count of the PV fault analysis papers that are associated to eight different PV fault parameters. Shading, short-circuit and degradation faults are researched the most while faulty conditions, line to line, soiling and open-circuit faults are less popular in PV fault research. The arc fault analysis using MLT present the least research papers.

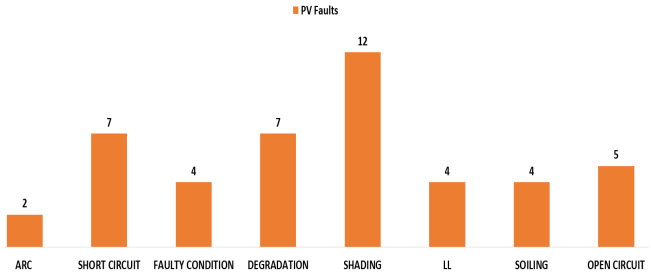


Fig. 7. PV System Faults

Table II presents some of the most popular model-based (MB) and data-driven regressive approaches (DDR) identified in the 90 papers studied in this work. The persistent and physical models were mostly used in the papers when using the model-based approach. While the ARIMA, Linear Model and PCA were mostly applied when using the data-driven regressive approach. The complete list of MB and DDR approaches can be found in the following reference [12].

TABLE II
LIST OF MODEL-BASED AND DATA-DRIVEN ALGORITHMS AND RESPECTIVE PAPER NUMBER COUNT

Model-Based Approach (14 experiments)	#	Data-Driven Regressive Approach (33 experiments)	#
Persistence Model	6	ARIMA (Auto-Regressive Integrated Moving Average)	7
Physical Model	2	Linear Model or Linear Regression	4
Hampel Identifier and Boxplot rule	1	PCA (Principal component analysis)	4

Table III presents all the MLT algorithms that were identified in the 90 papers studied in this work. The Random Forests algorithm is very popular amongst Tribe 1 MLTs followed by Regression Tree, Decision Tree, and Functional Tree algorithms. The ANN algorithm from Tribe 2 is highly used in the PVSM research field since it was identified 41 times followed by the ELM algorithm. Tribe 3 includes the Genetic Programming and Genetic algorithms which were not used much in the PVSM research. The Gaussian Process algorithm is the most popular MLT out of the Tribe 4 algorithms. The K-Nearest Neighbor algorithm is the most popular out of the Tribe 5 MLTs followed by the SVM, the SVR, SVR with Radial kernel, Radial Basis Function, Polynomial SVM or SVR, and the K-Means algorithm.

Table IV presents 4 of the 69 Hybrid MLT algorithms that

TABLE III
LIST OF MACHINE LEARNING ALGORITHMS ORGANIZED INTO FIVE MACHINE LEARNING TRIBES

Tribes	MLT	#
Tribe 1 # 15	Random Forests (RF)	4
	Regression Tree (RT)	2
	Classification and Regression Decision Tree (CART)	2
	Functional Tree	2
	C.45 Decision Tree	1
	Best First Decision Tree	1
	Gradient Boosting	1
	Binary Tree Bagging	1
	Fuzzy Logic (FL)	1
Tribe 2 # 58	Artificial Neural Network (ANN)	41
	Extreme Learning Machine (ELM)	4
	Extension Neural Network	2
	Recurrent Neural Network (RNN)	2
	Neural Network Ensemble (NNE) Bagging Or Bootstrap Based Neural Network (BNN)	2
	Group method of data handling neural network (GMDH-type NN)	1
	Dynamic Recurrent Neural Network (DRNN)	1
	Deep Belief Network (DBN)	1
	Long Short-Term Memory networks (LSTM) Based on Recurrent Neural Network (RNN)	1
	Auto-LSTM Long Short-Term Memory networks	1
	Entropy Extreme Learning Machine (EELM)	1
	Online Sequential Extreme Learning Machine	1
Tribe 3 #2	Genetic Programming (GP)	1
	Genetic Algorithm (GA)	1
Tribe 4 # 10	Gaussian Process	6
	Monte Carlo Method	1
	Graph-based Semi-Supervised Learning	1
	Naïve Bayes Classification	1
	Bayesian Belief Network	1
Tribe 5 # 15	K – Nearest Neighbor (K-NN)	9
	Support Vector Machine (SVM)	6
	Support Vector Regression (SVR)	5
	SVR with Radial kernel	4
	Radial Basis Function	3
	Polynomial SVM or SVR	2
	K-Means Algorithm	2
	Linear SVM	1
	Local Outlier Factor (LOF)	1
	Locally Weighted Learning	1
	Local Linear Fitting / Regression	1
	Clustering by fast search and find of density peaks	1

were identified in the 90 papers studied in this work. The RBFNN, the ANN+Physical model, the GA-ANN, and ANFIS were the most popular Hybrid MLTs. The complete list of the hybrid MLTs can be found in the following reference [12].

TABLE IV
LIST OF HYBRID MACHINE LEARNING ALGORITHMS

Hybrid MLT (69 algorithms)	**Tribes**	#
Radial Basis Function Neural Network (RBFNN)	T5 + T2	5
ANN + Physical Model	T2 + MB	5
Genetic Algorithm – ANN (GA- ANN)	T3 + T2	4
ANFIS (adaptive neuro-fuzzy inference system)	T1 + T2	3

A. Challenges and Opportunities for the PVSM research field

The MLTs suggested in Table V, present the best results comparison studies, however they have only been considered in between 1 and 3 experiment papers and therefore need further research. Based on the results, there are some opportunities and challenges that were identified to help further research in the PVSM field which are discussed as follows.

TABLE V
OPPORTUNITIES AND CHALLENGES

Opportunities	**Tribe 1** MLT present good results but few experiments, therefore should be further researched, particularly with the Random Forests, Decision Trees, Regression Trees and Fuzzy Logic.
	Tribe 2 presents a high experiment count due to the use of the ANN MLT, which indicates that the PVSM researchers are hesitant to try other MLT such as the ELM and RNN which have shown to present very accurate results.
	Only two research experiments have dealt with the **Tribe 3** (genetic programming and genetic algorithms) MLT.
	Tribe 4 MLT present very good results when using the Gaussian Process, the Monte Carlo and the Graph-based Semi-Supervised Learning. More research is encouraged.
	Amongst the **Tribe 5** MLTs, the K-NN MLT is more popular than the SVM, and therefore SVM work is encouraged.
	69 **Hybrid MLT** algorithms were identified but most explored only in one paper. More research should be done on combining MLT to discover the most accurate Hybrid MLTs.
	PV system **module-level monitoring** improves the accuracy of the results even though this solution is costly, however this should be further explored. PV fault **diagnostics and prognostics** research is initiating, but more emphasis should be put on PVSM prognostics to improve M&O scheduling and lower ROI payback times.
Challenges	There is a **vast number of MLT**s, therefore many researchers tend to use the same ones as they rely on research results from previous works and do not try new MLT.
	Research on PVSM should be done on real-world PV system **testbeds** to obtain credible results that consider real-world disturbances caused by environmental factors and signal noise coming from the measurement devices.

IV. CONCLUSION

This paper highlights the key parameters, challenges, and research opportunities that can further research in the MLT PVSM field. A list of MLTs organized by tribes is provided to narrow down the types of MLT that should be considered in future research. Hybrid MLT research and MLT comparison studies are encouraged. PV system Prognostics research is encouraged to provide condition-based maintenance scheduling and improve ROI payback times. PV module-level monitoring provides accurate results which is useful to explore PV faults. Real-world testbed research is encouraged, as accurate experimental results are presented.

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